

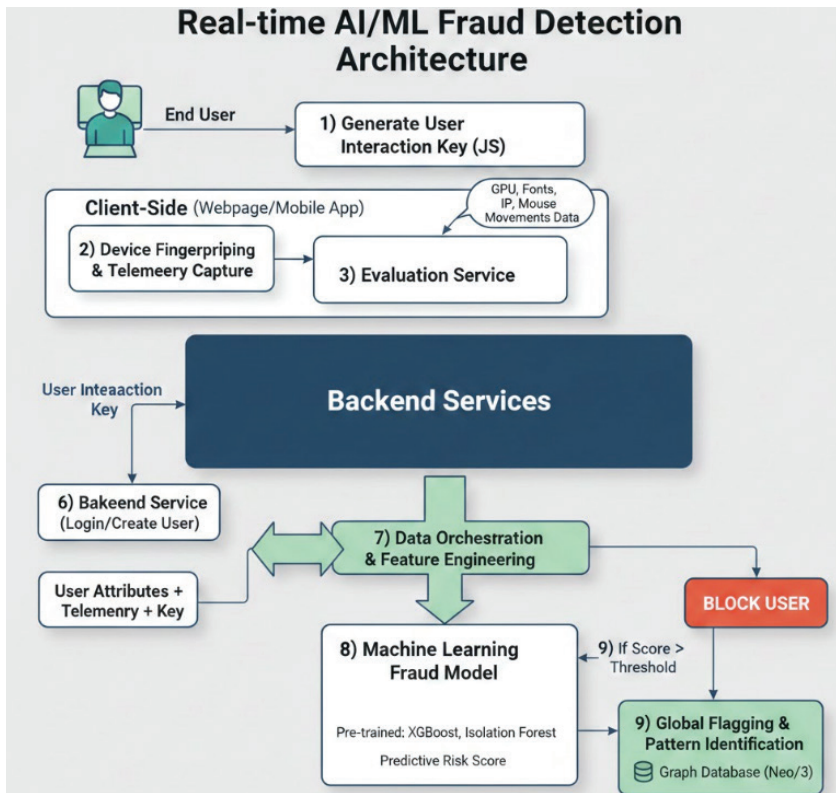


Figure 2: Architecture diagram

Phase II: Behavioural telemetry and interaction

- **Step 4: Continuous behavioural monitoring:** The system captures high-fidelity telemetry, including mouse trajectories, click patterns, and keystroke dynamics, which are buffered locally to build a behavioural biometric profile.
- **Step 5: Event-driven triggers:** The user initiates a critical action (e.g., account registration, authentication, or high-value downloads), serving as the catalyst for a risk evaluation.
- **Step 6: Payload aggregation and transmission:** The system bundles identity attributes (PII/form data) with the behavioural telemetry and the UIK, forwarding the unified payload to the relevant application API.

Phase III: Intelligent analysis and enforcement

- **Step 7: Multi-source data fusion:** The backend API retrieves the

stored device fingerprint and merges it with the behavioural and identity data, using the UIK as the join key to create a comprehensive feature vector.

- **Step 8: Machine learning inference:** This consolidated data is passed to a pre-trained predictive fraud model (e.g., XGBoost or Neural Network) to calculate a real-time risk score.
- **Step 9: Automated mitigation and global flagging: Real-time enforcement:** If the risk score exceeds the predefined threshold, the transaction is immediately blocked.
 - **Retrospective analysis:** The system performs a graph-based lookup to identify and flag historical activities or accounts associated with the same digital fingerprint, neutralising the fraud network at scale.

Behavioural digital fingerprinting as a detection strategy

Behavioural digital fingerprinting focuses on how interactions occur rather than who initiates them. Instead of relying solely on identity-centric signals, this approach analyses behavioural characteristics that are difficult to forge consistently at scale.

Examples of behavioural signals include:

- Keyboard interaction timing and cadence
- Navigation sequences and input patterns
- Request frequency and temporal distributions
- Cross-session behavioural consistency

When aggregated, these signals form behavioural fingerprints that capture interaction patterns unique to humans or automated systems. Unlike traditional identifiers, behavioural traits are harder to spoof consistently across devices and sessions, making them valuable for fraud detection.

Feature engineering from low-level interaction signals

Raw behavioural data is noisy and high-dimensional. Effective fraud detection requires transforming these signals into meaningful features that machine learning models can interpret.

Key feature categories include:

- **Temporal features:** Inter-keystroke intervals, session duration variability, and response timing distributions
- **Consistent features:** Repeated behavioural patterns across sessions or devices
- **Entropy measures:** Variability in input behaviour that distinguishes scripted automation from natural human interaction
- **Correlation features:** Similarity scores between behaviours observed across different identities or machines